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MACHINERY COST ESTIMATES

by

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The tables in this publication contain estimates of farm machinery operation costs calculated via an economic engineering approach. The data are intended to show a representative farming industry cost for specified machines and operations.

The 2024 machinery ownership and operating cost estimates are also available in a Microsoft Excel spreadsheet (MACHDATA.XLSM) which is available for downloading at **z.umn.edu/machdata**. Due to the wide variety of options and features in the farm equipment marketplace today it is becoming increasingly difficult to determine "typical" prices. While machine purchase prices are included here and in the spreadsheet, if possible I encourage you to enter your own purchase prices and the other input data in order to arrive at the most accurate results for your own situation.

Machine costs are separated into time-related and use-related categories. Use-related costs are incurred only when a machine is used. They include fuel, lubrication, use-related repairs and labor. Time-related costs, also often referred to as overhead costs, accrue to the owner whether or not a machine is used. Overhead includes time-related economic costs: interest, insurance, and housing. Depreciation is both a use- and a time-related cost. Depreciation will be related to use to the extent that increased annual usage shortens years of life and/or reduces salvage value. While not entirely use-related, depreciation is included along with operating expenses and labor costs in the columns labeled "use-related cost/acre".

OVERHEAD COSTS: Time-related costs are prorated over a 12-year economic life except where otherwise indicated. Trade-in values are estimated based on American Society of Agricultural Engineers formulas. Income tax implications are ignored. A housing charge of 97 cents per square foot of shelter space needed per year is made.

A five percent interest rate is used in the cost estimates. This is intended to approximate a "real" (inflation-adjusted) calculated by taking a nominal rate charged by lenders, minus a measure of the inflation rate per year expected over the years of ownership. Insurance is charged at 0.85 percent of the undepreciated value. The interest and insurance cost formulas

have one year's depreciation added to the numerator in effect bases the costs on the value at the beginning of each year owned. This gives a slightly more accurate calculation of the actual costs over the years owned. Salvage value are also adjusted for inflation so that depreciation is also on an inflation-adjusted basis. In states where farm machinery is taxed as personal property, property tax could be calculated in a similar manner, depending on how taxes are assessed.

Formulas used to compute machinery overhead costs:

$$\text{Interest, \$/year} = \frac{\text{purchase cost + salvage value + depreciation (\$/year)}}{2} \times \text{“real” interest rate}$$

$$\text{Insurance, \$/year} = \frac{\text{purchase cost + salvage value + depreciation (\$/year)}}{2} \times \text{insurance rate}$$

Housing, $/year = price per sq. foot x sq. feet shelter space required

USE-RELATED COSTS: Fuel cost is calculated by multiplying the fuel consumption by the price of fuel, with fuel consumption assumed to be 0.044 gallons of diesel fuel per PTO horsepower-hour on average for each implement type. Fuel consumption per acre is averaged across sizes within a given implement type. The price of farm diesel fuel is projected at $2.98 per gallon. All power units, tractors, combines, trucks, etc., use diesel fuel. Lubrication cost is assumed to be 10 percent of fuel cost.

The formulas for repair and maintenance costs estimate total accumulated repair costs based on accumulated hours of lifetime use. Repair and maintenance calculations are based on American Society of Agricultural Engineers formulas. The total cost is then divided by accumulated hours to arrive at an average per hour cost estimate. The amount of annual use of a machine is an estimate of the number of hours a commercial farmer would use that particular machine in one year.

Labor is charged at hourly rates that include benefits. The charge rates are $34.00 per hour for unskilled labor and $36.00 per hour for skilled labor. The skilled labor rate is generally used with the planting and harvesting equipment and sprayers. Labor per acre for an operation such as plowing or disking is calculated by using the work rate on the implement. Less labor per acre is used in a disking operation that covers more acres per hour than in a plowing operation. A small amount of extra labor is added over and above machine time to allow for downtime for tasks such as making adjustments and filling sprayers and planters. The labor adjustment ranges from 2 percent additional time for tillage to 25 percent for spraying.

Economic depreciation is included in use-related costs, calculated using the straight-line formula:

$$\text{Depreciation, \$/year} = \frac{\text{purchase cost - salvage value}}{\text{years you will use machine}}$$

These estimates will not represent any given individual's cost. Differences in buying power, repair programs, average annual use, and overall replacement programs should be considered when making adjustments. It may be useful to record actual expenses for at least a few of your implements and compare your costs to these estimates. These estimates will differ from records because they are estimates, but also because they are averaged over the use period and are expressed in today's dollars. If these estimates are compared to recorded costs that include repairs or depreciation based on historical costs, one adjustment that would be required for comparability would be to index the historical cost to current prices.

THE BOTTOM LINE: Machinery costs are substantial; control of them is important. Custom charges are often based upon them. No one should do custom work unless the charge will cover operating costs and use-related depreciation plus a return for one's risk and time. Ideally, all allocated per acre or hour overhead costs should also be covered by anyone offering to do custom work. The market for custom work usually does not cover all costs. The market is usually somewhere in between the use-related costs and total costs.

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Summary of Per Acre Use-Related Costs and Total Cost for Implements with Associated Power Units, Averaged Over All Sizes by Implement Type

	Use-Related Cost/Acre[1]	Total Cost/Acre
Chisel Plow	$14.05	$19.99
Chisel Plow, Front Dsk	$18.72	$25.73
Moldboard Plow	$27.15	$34.90
Field Cultivator	$7.07	$9.67
Tandem Disk	$15.26	$20.72
Row Crop Planter	$15.17	$22.30
Presswheel Drill	$15.80	$20.97
Air Seeder Drill w/Cart	$23.54	$33.60
No-Till Drill	$28.34	$38.22
Row Cultivator	$9.09	$12.02
Boom Sprayer, Self-Propelled	$6.73	$11.22
Boom Sprayer, Pull-Type	$4.71	$7.59
Stalk Shredder	$16.67	$21.88
Rotary Mower/Conditioner	$11.11	$14.87
Hay Rake	$2.82	$3.67
Hay Merger	$18.40	$21.15
Hay Merger, Self-Propelled	$14.05	$19.99
Hay Baler PTO Twine	$18.72	$25.73

	Use-Related Cost/Acre[1]	Total Cost/Acre
Round Baler w/Bale Wrap	$19.74	$22.94
Large Rectangular Baler	$15.66	$19.99
Forage Harvester, Pull-Type w/Corn Head	$75.24	$97.70
Forage Harvester, Pull-Type w/Pickup Head	$37.69	$48.29
Forage Harvester, Self-Prop Corn Head	$51.04	$69.65
Forage Harvester, Self-Prop Pickup Head	$54.97	$73.65
Forage Harvester, Self-Prop Pickup Head (2X windrows)	$33.18	$44.14
Combine Corn Hd	$48.01	$56.75
Combine Chopping Corn Hd	$52.26	$62.06
Combine Platform	$43.79	$51.01
Combine Belt Pickup Hd	$39.36	$47.01
Grain Cart	$32.33	$42.78

[1] Use-related cost/acre includes fuel, lubricants, repairs and maintenance, labor, and power and implement depreciation (depreciation is both time-related and use-related). The difference between use-related cost and total cost is that total cost also includes overhead costs (interest, insurance, and housing).

	Net Cost	Annual	Fuel & Oil	Maintenance	Depreciation	-- Overhead[3] --		Total Cost	Total Cost	Diesel
Tractor, combine or	of a New	Hours	Cost Per	& Repair	Cost Per	Cost Per	Cost Per	Per Year	Per Hour	Use/Hr
Forage Harvester HP[1]	Power Unit[2]	of Use	Hour	Cost/Hr	Hour	Year	Hour	Of Use	Of Use	Gallons
Tractors, Combines, and Self-Propelled Forage Harvesters (Without Heads)										
40 HP	$33,000	400	$5.77	$1.11	$4.02	$1,503	$3.76	$5,861	$14.65	1.76
60 HP	37,000	400	8.65	1.24	4.50	1,686	4.21	7,446	18.61	2.64
75 HP	65,000	400	10.82	2.30	7.62	2,934	7.33	11,226	28.07	3.30
105 HP MFWD	173,000	450	15.14	5.19	21.28	7,162	15.92	25,890	57.53	4.62
130 HP MFWD	239,000	450	18.75	7.17	29.40	9,850	21.89	34,746	77.21	5.72
160 HP MFWD	299,000	500	23.08	9.97	33.34	12,321	24.64	45,515	91.03	7.04
200 HP MFWD	387,000	500	28.85	12.90	43.16	15,900	31.80	58,352	116.70	8.80
225 HP MFWD	402,000	400	32.45	10.72	55.22	16,653	41.63	56,009	140.02	9.90
260 HP MFWD	480,000	400	37.50	7.68	65.93	19,837	49.59	64,282	160.71	11.44
310 HP 4WD	491,000	400	44.71	7.86	67.44	20,286	50.71	68,291	170.73	13.64
360 HP 4WD	451,000	400	51.92	7.22	61.95	18,653	46.63	67,089	167.72	15.84
425 HP 4WD	536,000	400	61.30	8.58	73.63	22,123	55.31	79,523	198.81	18.70
275 HP Combine	447,000	400	39.66	105.71	82.32	15,943	39.86	107,018	267.54	12.10
375 HP Combine	525,000	400	54.09	124.15	96.68	18,754	46.89	128,722	321.80	16.50
440 HP Combine	557,000	400	55.25	131.72	102.57	19,868	49.67	135,684	339.21	16.85
400 HP SP Forage Harvester Base Unit	524,000	350	31.47	73.19	68.46	23,429	66.94	84,021	240.06	9.60
625 HP SP Forage Harvester Base Unit	630,000	350	49.17	88.00	82.31	28,071	80.20	104,888	299.68	15.00

[1]HP shown for the smaller tractors is PTO horsepower. Engine HP is shown for the larger tractors. PTO HP for the larger tractors runs about 87% of engine HP, and is shown in parentheses. Fuel use is estimated at 0.044 gallons of diesel fuel per hour per PTO HP.

[2]Net cost of a new unit assumes no trade-in. Farm machinery is exempt from sales tax in Minnesota so no sales tax is included.

[3]Overhead costs include interest, insurance, and housing but not depreciation, which is shown separately because it varies to some extent with use. Overhead pe hour will vary with annual use.

	Tractor	Net Cost	--Estimated--				Labor	--Implement Cost/Acre--			Total	Use-related	Operating	Diesel
	Size	of a New	Work-Performed		-Power Cost/Acre[2]-		Cost		Deprec-		Cost	Cost	Cost	Fuel
Implement	(HP)	Implement[1]	Acres/hr	Acres/yr	Fuel	Other	Per Acre	Repairs	iation	Overhead[3]	Per Acre[4]	Per Acre[5]	Per Acre	Gal/Acre
Tillage														
Chisel Plow 23 Ft	200 HP MFWD	$59,500	11.85	711	$2.14	$7.71	$2.93	$1.37	$4.79	$3.69	$ 22.64	$ 16.26	$ 7.67	0.70
Chisel Plow 37 Ft	310 HP 4WD	$87,000	19.06	1,144	$2.23	$6.72	$1.82	$1.25	$4.36	$3.36	$ 19.74	$ 13.72	$ 5.77	0.70
Chisel Plow 57 Ft	425 HP 4WD	$146,000	29.36	1,762	$2.49	$4.28	$1.18	$1.36	$4.75	$3.52	$ 17.58	$ 12.17	$ 5.12	0.70
Chisel Plow, Front Dsk 16.3 Ft	200 HP MFWD	$38,000	8.37	670	$4.08	$9.86	$4.14	$0.88	$3.22	$2.61	$ 24.80	$ 18.39	$ 10.33	1.15
Chisel Plow, Front Dsk 21.3 Ft Fold	310 HP 4WD	$59,500	10.95	876	$3.45	$12.15	$3.17	$1.06	$3.86	$2.98	$ 26.66	$ 19.05	$ 8.71	1.15
Moldboard Plow 6 Bottom-18, 9 Ft	130 HP MFWD	$33,500	4.17	501	$4.15	$14.36	$8.31	$3.46	$3.83	$2.94	$ 37.05	$ 28.87	$ 17.81	1.32
Moldboard Plow 8 Bottom-18, 12 Ft	160 HP MFWD	$44,500	5.56	668	$4.49	$11.87	$6.23	$3.45	$3.82	$2.89	$ 32.75	$ 25.43	$ 15.80	1.32
Field Cultivator 23 Ft	105 HP MFWD	$54,000	16.59	1,659	$1.12	$2.35	$2.09	$1.05	$1.86	$1.42	$ 9.90	$ 7.52	$ 4.47	0.31
Field Cultivator 47 Ft	260 HP MFWD	$111,000	33.90	3,390	$0.93	$3.81	$1.02	$1.06	$1.88	$1.43	$ 10.12	$ 7.23	$ 3.32	0.31
Field Cultivator 60 Ft	310 HP 4WD	$139,000	43.27	4,327	$1.00	$2.94	$0.80	$1.04	$1.84	$1.38	$ 9.00	$ 6.45	$ 3.04	0.31
Tandem Disk 21 Ft Fold	160 HP MFWD	$89,000	12.22	1,222	$2.97	$4.48	$2.84	$2.44	$4.17	$3.12	$ 20.02	$ 14.88	$ 8.52	0.74
Tandem Disk 30 Ft Fold	360 HP 4WD	$129,000	17.45	1,745	$1.89	$7.72	$1.99	$2.47	$4.23	$3.12	$ 21.42	$ 15.63	$ 7.31	0.74
Planting														
Row Crop Planter 6 Row-30, 15 Ft	60 HP	$47,500	7.00	490	$1.41	$1.25	$5.97	$1.99	$4.42	$4.64	$ 19.68	$ 14.43	$ 9.46	0.40
Row Crop Planter 8 Row-30, 20 Ft	75 HP	$59,000	9.33	653	$1.49	$1.52	$4.47	$1.86	$4.12	$4.26	$ 17.71	$ 12.67	$ 7.90	0.40
Row Crop Planter 12 Row-30, 30 Ft	105 HP MFWD	$136,000	14.00	980	$1.57	$2.54	$2.98	$2.85	$6.33	$6.38	$ 22.65	$ 15.13	$ 7.53	0.40
Row Crop Planter 16 Row-30, 40 Ft	200 HP MFWD	$200,000	18.67	1,307	$1.10	$5.15	$2.24	$3.14	$6.98	$6.93	$ 25.54	$ 16.91	$ 7.40	0.40
Row Crop Planter 24 Row-30, 60 Ft	310 HP 4WD	$324,000	28.00	1,960	$1.05	$5.05	$1.49	$3.40	$7.53	$7.39	$ 25.91	$ 16.71	$ 6.49	0.40
Presswheel Drill 16 Ft	105 HP MFWD	$29,500	6.79	509	$1.78	$6.70	$5.89	$1.38	$3.42	$2.58	$ 21.73	$ 16.81	$ 10.03	0.61
Presswheel Drill 20 Ft	130 HP MFWD	$35,000	8.48	636	$1.80	$7.30	$4.71	$1.31	$3.24	$2.46	$ 20.81	$ 15.78	$ 8.87	0.61
Presswheel Drill 25 Ft	130 HP MFWD	$63,000	10.61	795	$2.24	$5.04	$3.77	$1.88	$4.67	$3.41	$ 21.01	$ 15.53	$ 8.33	0.61
Presswheel Drill 30 Ft	160 HP MFWD	$79,500	12.73	1,018	$2.20	$4.95	$3.14	$2.12	$4.60	$3.32	$ 20.33	$ 15.08	$ 8.05	0.61
Air Seeder Drill w/Cart 52 Ft	260 HP MFWD	$342,000	22.06	1,765	$1.70	$5.58	$1.81	$5.27	$11.42	$7.81	$ 33.60	$ 23.54	$ 9.13	0.52
No-Till Drill 15 Ft	130 HP MFWD	$79,000	6.36	509	$2.95	$9.19	$6.28	$4.22	$9.15	$6.44	$ 38.22	$ 28.34	$ 14.57	0.90
Crop Maintenance														
Row Cultivator 12 Row-30, 30 Ft	160 HP MFWD	$48,000	15.45	1,545	$1.49	$4.40	$2.29	$0.73	$1.78	$1.33	$ 12.02	$ 9.09	$ 5.16	0.46
Boom Sprayer, Self-Propelled 120 Ft	None	$673,000	66.18	6,618	$0.45	-$0.45	$0.68	$1.19	$4.63	$4.49	$ 11.22	$ 6.73	$ 2.10	0.07
Boom Sprayer, Pull-Type 90 Ft	130 HP MFWD	$54,500	46.09	1,152	$0.41	$1.27	$0.98	$0.38	$2.16	$2.41	$ 7.59	$ 4.71	$ 1.91	0.12

	Tractor	Net Cost	--Estimated--				Labor	--Implement Cost/Acre--			Total	Use-related	Operating	Diesel
	Size	of a New	Work-Performed		-Power Cost/Acre[2]-		Cost		Deprec-		Cost	Cost	Cost	Fuel
Implement	(HP)	Implement[1]	Acres/hr	Acres/yr	Fuel	Other	Per Acre	Repairs	iation	Overhead[3]	Per Acre[4]	Per Acre[5]	Per Acre	Gal/Acre
Stalk Shredder 20 Ft	130 HP MFWD	$40,500	7.76	776	$2.42	$7.54	$4.82	$1.75	$2.97	$2.39	$ 21.88	$ 16.67	$ 9.91	0.74
Harvest														
Rotary Mower/Conditioner 12 Ft	75 HP	$47,500	8.73	698	$1.24	$1.98	$4.09	$0.93	$3.71	$2.92	$ 14.87	$ 11.11	$ 6.52	0.38
Hay Rake 30 Ft	40 HP	$31,000	26.18	2,095	$0.22	$0.34	$1.32	$0.22	$0.86	$0.70	$ 3.67	$ 2.82	$ 1.81	0.07
Hay Merger 14 Ft	75 HP	$74,000	10.86	2,172	$0.87	$1.71	$3.19	$1.83	$1.99	$1.51	$ 11.10	$ 8.92	$ 6.17	0.29
Hay Merger 34 Ft	160 HP MFWD	$295,000	26.38	5,275	$1.00	$2.46	$1.31	$1.79	$3.05	$2.35	$ 11.96	$ 8.67	$ 4.42	0.29
Hay Baler PTO Twine 12 Ft	40 HP	$37,000	4.36	873	$1.32	$2.04	$9.16	$4.28	$2.47	$1.89	$ 21.15	$ 18.40	$ 15.01	0.40
Round Baler w/Bale Wrap 5x6 , 20 Ft	75 HP	$114,000	9.45	1,891	$1.14	$1.82	$3.67	$10.36	$3.52	$2.43	$ 22.94	$ 19.74	$ 15.41	0.35
Large Rectangular Baler 3x3 , 20 Ft	130 HP MFWD	$155,000	11.64	2,909	$1.61	$5.02	$3.43	$3.56	$3.11	$2.20	$ 18.94	$ 14.86	$ 9.23	0.49
Large Rectangular Baler 4x3 , 20 Ft	130 HP MFWD	$192,000	11.64	2,909	$1.61	$5.02	$3.43	$4.42	$3.85	$2.71	$ 21.04	$ 16.45	$ 10.08	0.49
Forage Harvester, Pull-Type w/Corn Head 3 Row, 7.5 Ft	160 HP MFWD	$96,500	2.07	414	$11.16	$32.86	$19.32	$13.15	$10.67	$10.54	$ 97.70	$ 75.24	$ 48.45	3.40
Forage Harvester, Pull-Type w/Pickup Head 12 Ft	105 HP MFWD	$84,500	3.31	662	$4.58	$12.81	$12.08	$7.20	$5.84	$5.80	$ 48.29	$ 37.69	$ 25.42	1.40
Forage Harvester, Self-Prop Corn Head 6 Row, 15 Ft	625 HP SP Forage Harvester Base Unit	$107,000	5.09	1,018	$7.24	$51.62	$7.85	$1.68	$4.81	$4.89	$78.09	$ 57.45	$ 35.27	2.58
Forage Harvester, Self-Prop Corn Head 8 Row, 20 Ft	625 HP SP Forage Harvester Base Unit	$141,000	6.79	1,358	$9.66	$34.49	$5.89	$1.66	$4.75	$4.76	$61.21	$ 44.63	$ 28.96	2.58
Forage Harvester, Self-Prop Pickup Head 12 Ft	400 HP SP Forage Harvester Base Unit	$35,000	4.07	815	$7.73	$51.22	$9.81	$0.69	$1.96	$2.24	$73.65	$ 54.97	$ 36.20	2.36
Forage Harvester, Self-Prop Pickup Head (2X windrows) 24 Ft	625 HP SP Forage Harvester Base Unit	$35,000	8.15	1,629	$6.04	$30.75	$4.91	$0.34	$0.98	$1.12	$44.14	$ 33.18	$ 22.09	1.84
Combine Platform 20 Ft	275 HP Combine	$36,000	5.94	1,188	$6.68	$38.37	$6.73	$0.79	$2.14	$1.17	$55.88	$ 48.00	$ 32.00	2.04
Combine Platform 25 Ft	375 HP Combine	$43,500	7.42	1,485	$6.07	$37.27	$5.38	$0.77	$2.07	$1.12	$52.68	$ 45.25	$ 29.55	2.04
Combine Platform 30 Ft	375 HP Combine	$51,000	8.91	1,782	$7.29	$28.84	$4.49	$0.75	$2.02	$1.08	$44.46	$ 38.11	$ 25.85	2.04
Combine Corn Hd 6 Row-30, 15 Ft	275 HP Combine	$66,000	5.09	1,018	$5.30	$47.25	$7.85	$1.70	$4.58	$2.43	$69.11	$ 58.85	$ 36.86	2.00
Combine Corn Hd 8 Row-30, 20 Ft	275 HP Combine	$85,500	6.79	1,358	$7.25	$32.16	$5.89	$1.65	$4.45	$2.37	$53.77	$ 45.53	$ 29.66	2.00
Combine Chopping Corn Hd 8 Row-30, 20 Ft	275 HP Combine	$115,000	6.79	1,358	$6.60	$32.81	$5.89	$7.65	$4.53	$3.61	$61.10	$ 51.61	$ 35.34	1.90
Combine Corn Hd 12 Row-30, 30 Ft	375 HP Combine	$131,000	10.18	2,036	$7.78	$23.82	$3.92	$1.69	$4.54	$2.38	$44.13	$ 37.15	$ 24.35	2.00
Combine Corn Hd 12 Row-22, 22 Ft	375 HP Combine	$129,000	7.47	1,493	$5.85	$37.25	$5.35	$2.26	$6.10	$3.20	$60.01	$ 50.53	$ 30.79	2.00
Combine Chopping Corn Hd 12 Row-30, 30 Ft	440 HP Combine	$174,000	10.18	2,036	$7.02	$26.30	$3.92	$7.72	$4.57	$3.60	$53.14	$ 44.65	$ 30.80	1.90

Implement	Tractor Size (HP)	Net Cost of a New Implement[1]	--Estimated-- Work-Performed Acres/hr	Acres/yr	-Power Cost/Acre[2]- Fuel	Other	Labor Cost Per Acre	--Implement Cost/Acre-- Repairs	Deprec-iation	Overhead[3]	Total Cost Per Acre[4]	Use-related Cost Per Acre[5]	Operating Cost Per Acre	Diesel Fuel Gal/Acre
Combine Chopping Corn Hd 12 Row-22, 22 Ft	440 HP Combine	$170,000	7.47	1,493	$5.05	$40.38	$5.35	$10.29	$6.09	$4.80	$71.96	$ 60.50	$ 39.50	1.90
Combine Belt Pickup Hd 23 Ft	275 HP Combine	$56,000	7.81	832	$5.08	$29.19	$5.12	$0.47	$4.60	$2.55	$47.01	$ 39.36	$ 24.21	1.55
Grain Cart 30 Ft	225 HP MFWD	$152,000	6.87	1,375	$4.72	$15.65	$5.49	$6.07	$6.45	$4.39	$42.78	$ 32.33	$ 17.84	1.44

[1]Net cost of a new unit assumes no trade-in. Farm machinery is exempt from sales tax in Minnesota so no sales tax is included.

[2]Power cost per acre for the power unit assigned to each implement multiplied times that implement's acres/hour equals that power unit's total cost per hour shown in the "Tractors, Combines, and Self-Propelled Forage Harvesters (Without Heads)" table above.

[3]Overhead per acre will vary with annual use.

[4]Total cost/acre is total cost per hour divided by acres per hour. Includes fuel, lubricants, power and equipment repairs and maintenance, labor, and overhead costs including depreciation. Fuel is included in power cost.

[5]Use-related cost/acre is included in the total cost/acre amount. Use-related cost/acre includes everything in total cost/acre EXCEPT that non-depreciation overhead costs (interest, insurance, and housing) are omitted. Depreciation is included in use-related cost under the assumption that extra use reduces trade-in value which increases annual depreciation. In other words, depreciation is considered here to be at least partially use-related even though it is commonly thought of as being mainly time-related.